



Title

rdsensitivity — Sensitivity analysis for RD designs under local randomization.

Syntax

```
rdsensitivity outvar runvar [if] [in] [, cutoff(#) wlist(numlist)
wlist_left(numlist) tlist(numlist) statistic(stat_name) p(#) vce(hc1|hc2|hc3)
evalat(point) kernel(kerneltype) fuzzy(fuzzy_var [fuzzy_stat]) ci(# #)
ci_alpha(#) reps(#) seed(#) saving(filename) nodots nodraw verbose ]
```

Description

rdsensitivity analyzes the sensitivity of randomization p-values and confidence intervals to different window lengths for regression discontinuity (RD) designs under local randomization. See [Cattaneo, Frandsen and Titiunik \(2015\)](#) and [Cattaneo, Titiunik and Vazquez-Bare \(2017\)](#) for an introduction to this methodology.

A detailed introduction to this command is given in [Cattaneo, Titiunik and Vazquez-Bare \(2016\)](#). Companion R and Python packages are also available [here](#).

Companion functions are [rdrandinf](#), [rdwinselect](#) and [rdrbounds](#).

Related R, Python, and Stata packages useful for inference in RD designs are described in the following website:

<https://rdpackages.github.io/>

With **p()** greater than zero, the grid contains large-sample normal p-values for mean contrasts or fuzzy Anderson-Rubin tests, with HC3 standard errors by default. Supply **tlist** explicitly; confidence sets invert these tests over the supplied grid. For **ksmirnov** or **ranksum**, a positive **p** is replaced by **p(0)** and a warning is issued at the end. Inference with **p(0)** is unchanged.

Options

cutoff(#) specifies the RD cutoff for the running variable *runvar*. Default is **cutoff(0)**.

_____ Window and TE lists _____

wlist(*numlist*) specifies the list of window limits to the right of the cutoff. By default the program constructs 10 symmetric windows around the cutoff, the first one including 10 treated and control observations and then adding 5 observations to each group in subsequent windows.

wlist_left(*numlist*) specifies the list of window limits to the left of the cutoff. When not specified, the command uses symmetric windows around the cutoff based on **wlist()**.

tlist(*numlist*) specifies the list of null values for the treatment effect. By default the program uses ten evenly spaced points within the asymptotic confidence interval for a constant treatment effect in the smallest window to be used.

_____ Statistic _____

statistic(*stat_name*) specifies the statistic to be used. Options are:
diffmeans for difference in means statistic. This is the default option.
ksmirnov for Kolmogorov-Smirnov statistic.
ranksum for Wilcoxon-Mann-Whitney studentized statistic.
The option **ttest** is equivalent to **diffmeans** and included for backward compatibility.

p(#) specifies the order of the polynomial for outcome adjustment model. Default is **p(0)**.

vce(*hc1/hc2/hc3*) selects the heteroskedasticity-consistent variance estimator for **p()** greater than zero. Default is **vce(hc3)**. Ignored when **p(0)**.

evalat(*point*) specifies the point at which the adjusted variable is evaluated. Allowed options are **cutoff** and **means**. Default is **evalat(cutoff)**.

kernel(*kerneltype*) specifies the type of kernel to use as a weighting scheme. Allowed kernel types are **uniform** (uniform kernel), **triangular** (triangular kernel), and **epan** (Epanechnikov kernel). Default is **kernel(uniform)**.

fuzzy(*fuzzy_var [fuzzy_stat]*) specifies the endogenous treatment variable in a fuzzy design. This option uses an Anderson-Rubin/intention-to-treat statistic.

Inference

ci(# #) returns the confidence interval corresponding to the indicated window. The first value is the left limit and the second value is the right limit of the window. The values in **ci()** must match one of the windows defined by **wlist()** and **wlist_left()**.

ci_alpha(#) specifies the value of alpha for the confidence interval. Default alpha is 0.05, corresponding to a 95 percent confidence interval.

reps(#) specifies the number of replications. Default is **reps(1000)**.

seed(#) sets the seed for the randomization test. With this option, the user can manually set the desired seed, or can enter the value -1 to use the system seed. Default is **seed(666)**.

Save and display output

saving(*filename*) saves the dataset containing the data for the contour plot in *filename*. This allows the user to replicate and modify the appearance of the plot, and also conduct further sensitivity analysis.

nodots suppresses replication dots.

nodraw suppresses contour plot.

verbose displays matrix of results.

Example: Cattaneo, Frandsen and Titiunik (2015) Incumbency Data

```

Setup
    . use rdlocrand_senate.dta, clear

Sensitivity analysis using 1000 replications
    . rdsensitivity demvoteshfor2 demmv, wlist(.75(.25)2) tlist(0(1)20) reps(1000)

Obtain confidence interval for window [-.75;.75]
    . rdsensitivity demvoteshfor2 demmv, wlist(.75(.25)2) tlist(0(1)20) reps(1000)
    ci(-.75 .75)

```

```
Replicate contour graph using saved dataset
. rdsensitivity demvoteshfor2 demmv, wlist(.75(.25)2) tlist(0(1)20) reps(1000)
  saving(graphdata)
. use graphdata, clear
. twoway contour pvalue t w, ccuts(0(0.05)1)
```

Saved results

rdsensitivity saves the following in **r()**:

Scalars

r(p_requested)	requested polynomial degree
r(p)	effective polynomial degree

Locals

r(vce)	HC estimator when the effective polynomial degree is positive
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Matrices

r(results)	matrix of p-values.
r(CI)	confidence interval matrix (if ci option is specified).

References

- Cattaneo, M. D., Frandsen, B., and R. Titiunik. 2015. Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate. *Journal of Causal Inference* 3(1): 1-24.
- Cattaneo, M.D., Titiunik, R. and G. Vazquez-Bare. 2016. Inference in Regression Discontinuity Designs under Local Randomization. *Stata Journal* 16(2): 331-367.
- Cattaneo, M. D., Titiunik, R. and G. Vazquez-Bare. 2017. Comparing Inference Approaches for RD Designs: A Reexamination of the Effect of Head Start on Child Mortality. *Journal of Policy Analysis and Management* 36(3): 643-681.

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